

Equivalent Fractions and Comparing — Look at Both Numbers

Answer Key

Grades 4–5

Quick Answers

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|-------|---|
| 1. — | 5. —, 21 |
| 2. — | 6. $\frac{3}{4}$ |
| 3. — | 7. $\frac{3}{5}$, $\frac{2}{3}$, $\frac{7}{10}$ |
| 4. 24 | 8. —, 15 |
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Curriculum

Level: Grades 4–5

3.NF.A / 4.NF.A – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 11 tagged checks.

Common wrong answers

4. *If they got 3: multiplied only the bottom number and left the top one alone*
5. *If they got 3: kept the top number when only the bottom was multiplied up to twenty-eight*
8. *If they got 5: wrote the top number again after multiplying only the bottom up to eighteen*
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1. Compare $\frac{3}{8}$ and $\frac{5}{8}$.

Step 1. Check what matches. Both bottoms are 8, so both fractions are built from eighths — pieces of exactly the same size.

Step 2. When the pieces are the same size, counting them is a fair comparison: three eighths against five eighths.

Step 3. Five pieces beat three pieces, so $\frac{3}{8} < \frac{5}{8}$

Step 4. Why the sentence matters: this is the ONE case where looking at the top numbers alone is safe, and knowing why keeps it from being used everywhere else.

2. Compare $\frac{3}{5}$ and $\frac{3}{8}$.

Step 1. Check what matches. This time the TOPS match: three pieces in each fraction. The bottoms differ, so the pieces are different sizes.

Step 2. Cutting a whole into 8 pieces makes smaller pieces than cutting it into 5. So a fifth is bigger than an eighth.

Step 3. Three of the bigger pieces beat three of the smaller ones: $\frac{3}{5} > \frac{3}{8}$

Step 4. This is the pattern that trips students up: with the same top number, the fraction with the BIGGER bottom number is the SMALLER fraction.

3. Ben: $\frac{2}{9} > \frac{2}{4}$ because “nine is bigger than four”.

Step 1. Name the error. Ben compared the bottom numbers as if they were amounts. A bottom number is not an amount — it counts how many pieces the whole was cut into.

Step 2. More pieces means each piece is smaller, so a ninth is smaller than a quarter.

Step 3. Both fractions hold two pieces, so the one made of smaller pieces is less: $\frac{2}{9} < \frac{2}{4}$

Step 4. A picture settles it for a doubtful student: two slices of a pizza cut into nine is clearly less food than two slices of the same pizza cut into four.

4. Cara: $\frac{3}{5}$ rewritten as $\frac{3}{40}$.

Step 1. Find what was done to the bottom: 5 became 40, and $40 \div 5 = 8$, so the bottom was multiplied by 8.

Step 2. An equivalent fraction needs the SAME multiplication on top. Cara did it to only one number, which does not make an equal fraction — it makes a completely different one.

Step 3. Multiply the top by 8 as well: $3 \times 8 = 24$, so the correct top number is $\boxed{24}$ and $\frac{3}{5} = \frac{24}{40}$.

Step 4. Check the size: $\frac{3}{5}$ is a little more than half, and $\frac{24}{40}$ is a little more than half too.

Cara’s $\frac{3}{40}$ is nowhere near half, which is the quickest way to catch this error.

5. Which is greater, $\frac{3}{4}$ or $\frac{5}{7}$? Rewrite both with 28 on the bottom.

Step 1. Neither the tops nor the bottoms match, so no shortcut applies. Find a bottom number both can be rewritten with: $4 \times 7 = 28$ works.

Step 2. Rewrite the first fraction. The bottom went from 4 to 28, a multiplication by 7, so the top does the same: $3 \times 7 = 21$, giving $\frac{21}{28}$. The top number is $\boxed{21}$

Step 3. Rewrite the second. Its bottom went from 7 to 28, a multiplication by 4, so $5 \times 4 = 20$, giving $\frac{20}{28}$.

Step 4. Now the pieces are the same size, so count them: 21 beats 20, and $\frac{3}{4} > \frac{5}{7}$

Step 5. Notice what the shortcut would have said: 5 is bigger than 3, so a top-numbers-only reader picks the wrong fraction — by a single twenty-eighth.

6. Write $\frac{6}{8}$ in simplest form.

Step 1. Look for a number that divides both the top and the bottom. Both 6 and 8 are even, so 2 works.

Step 2. Divide both by the same number: $6 \div 2 = 3$ and $8 \div 2 = 4$.

Step 3. $\frac{3}{4}$ — and nothing divides 3 and 4 together except 1, so this is as simple as it gets.

Step 4. Same rule as problem 4, run backwards: equivalent fractions come from doing the same thing to BOTH numbers, whether that is multiplying or dividing.

7. Order $\frac{2}{3}$, $\frac{3}{5}$, $\frac{7}{10}$ from smallest to largest.

Step 1. The bottoms are 3, 5 and 10, and all three divide into 30, so rewrite everything in thirtieths.

Step 2. $\frac{2}{3} = \frac{20}{30}$ (top and bottom $\times 10$); $\frac{3}{5} = \frac{18}{30}$ ($\times 6$); $\frac{7}{10} = \frac{21}{30}$ ($\times 3$).

Step 3. Same-sized pieces at last, so order the tops: 18, then 20, then 21.

Step 4. Written as the original fractions: $\frac{3}{5} < \frac{2}{3} < \frac{7}{10}$

Step 5. A top-numbers-only reader would answer $2/3$, $3/5$, $7/10$ — wrong at the very first step — and a bottom-numbers-only reader would answer $7/10$, $3/5$, $2/3$, which is exactly backwards.

8. (a) Why “just look at the top numbers” fails. (b) Compare $\frac{5}{6}$ and $\frac{7}{9}$ over 18.

(a) *Open response — flagged for manual review. Model answer below; accept any equivalent reasoning with a working example.*

The top number only counts pieces; it says nothing about how big a piece is. The bottom number sets the size, and more pieces means smaller ones. So two fractions can only be compared by their tops when their bottoms already match. Example: $\frac{2}{9}$ and $\frac{2}{4}$ have the

same top, so the shortcut calls them equal, but $\frac{2}{4}$ is a half and $\frac{2}{9}$ is much less. Any example with a matching or larger top over a much larger bottom works.

What a full-credit answer must contain: the idea that the bottom number sets the size of the pieces, and one example where the shortcut gives the wrong answer.

(b) Step 1. Neither number matches, and both 6 and 9 divide into 18, so rewrite both fractions in eighteenths.

Step 2. The first bottom goes from 6 to 18, a multiplication by 3, so the top does the same: $5 \times 3 = 15$. The top number is $\boxed{15}$

Step 3. The second bottom goes from 9 to 18, a multiplication by 2, so $7 \times 2 = 14$, giving $\frac{14}{18}$.

Step 4. Fifteen eighteenths beats fourteen eighteenths, so $\frac{5}{6} > \frac{7}{9}$

Step 5. The shortcut fails here in both directions: the tops say $7 > 5$ and the bottoms say $9 > 6$, and each “reason” points at the smaller fraction.